# Writ

Physically settled BTC options, fully collateralized, on GIWA

| | |
|---|---|
| **Live demo** | `https://www.writ.fi` |
| **Source code** | `https://github.com/erenyegit/writ2` |
| **Network** | `GIWA Sepolia · chain 91342` |

Writ is an options desk built for income rather than leverage. A user sells a cash-secured put or a covered call and **receives the premium immediately**. Every position is collateralized at its full notional, so there are **no liquidations, no margin calls and no pooled risk** between writers. Expiry settles against the first Pyth price at or after the expiry timestamp, which no party can influence by choosing when to claim.

## Two products

| Product | Writer locks | If the price comes |
|---|---|---|
| **Cash-secured put** | `strike × size` USDC | buys BTC at the strike |
| **Covered call** | `size` in BTC | sells BTC at the strike |

## Physical settlement is the product

A cash-settled option pays a difference, which is not what the writer asked for. Someone who says *"I would buy BTC at $74,000"* and sees it finish at $70,000 gets a smaller balance from a cash-settled desk. Writ gives them **BTC, at $74,000** — the price they named, plus the premium they were already paid.

At expiry an option is either out of the money, and every leg returns to where it came from, or assigned, and the two sides swap **in full** at the strike. There is no partial payout, because no difference is being settled.

## A venue, not a house

- Makers hold their own balances in the contract and sign their own quotes. The protocol matches and settles; it never takes a side.
- Writing an option **reserves the side that maker may owe**, so no position can open that its counterparty could not honour at expiry.
- A quote names its maker and is checked against **that maker's** key, so one maker's signer can never commit another's inventory.
- A request for quote asks every registered maker at once, holds a short window open and returns the best bid. A maker that misses the window is skipped rather than closing the market.

## Deployed contracts · GIWA Sepolia

| | |
|---|---|
| **WritOptions** | `0xca0470B69c5fb9f28C64d65228f3e4879b85b4F2` |
| **Pyth oracle adapter** | `0x3B47e405e8195F5528958f568a48c6C723a9E5BD` |
| **TestUSDC** | `0x0b76264Bb1eD80e5FB219828464d0C809567e309` |

| | |
|---|---|
| **TestBTC** | `0xD23c5fA0f4029874be6338fdaDC66e81709DAbFf` |
| **Pyth** | `0x2880aB155794e7179c9eE2e38200202908C17B43` |

- All four of our contracts are **verified with source** on sepolia-explorer.giwa.io.
- GIWA has neither a canonical stablecoin nor a wrapped BTC, so both settlement assets are faucet-backed testnet tokens. Every amount and formula carries over unchanged to bridged assets on mainnet.

## Properties enforced by the contract

- **Solvency, per asset.** Each token balance always equals `makerFree + makerReserved + writerCollateral`. Both are fuzz-tested.
- **Delivery is pre-funded.** A position cannot be opened unless its counterparty already holds what it might owe, so assignment can never fail for lack of inventory.
- **A quote signer cannot move funds.** It only sets premiums on its own maker's quotes.
- **Reserves are unreachable.** A maker may withdraw only its uncommitted balance.
- **Settlement cannot be timed.** `parsePriceFeedUpdatesUnique` pins the first tick after expiry, with confidence-interval and exponent guards.
- **The oracle sits behind an adapter.** `IOracleAdapter` lets a second feed drop in without touching the core.

WHY GIWA

### The same desk returns more premium on GIWA.

Writ runs on signed quotes. That design has exactly two weaknesses, and GIWA removes both at the architecture level rather than by convention.

**Preconfirmation.** A quote must survive until the writer's transaction lands. The longer it lives, the more stale-price risk the maker prices into the spread. Flashblocks preconfirm in roughly 200 ms, so our quotes live **20 seconds instead of 60**: about three times less stale risk, and correspondingly more premium kept by the writer.

**Private orderflow.** There is no public mempool. Only the sequencer sees pending transactions, so **a signed quote cannot be observed and traded against before it lands**.

## Live on-chain proof

| Metric | Value |
|---|---|
| **Tests passing** | 83, including settlement against real Pyth updates and a solvency check per collateral asset |
| **Maker funded** | `2,000.000000 USDC · 1.00000000 BTC` |
| **First position** | Put, strike $74,000, size 0.001 BTC, 3-day expiry |
| **Collateral locked** | `74.000000 USDC` |
| **Premium paid upfront** | `0.473958 USDC` |
| **Delivery reserved** | `0.00100000 BTC, held against assignment` |

Both invariants hold exactly on-chain: `1,873.526042 = 1,799.526042 + 0 + 74.000000` and `1.00000000 = 0.99900000 + 0.00100000 + 0`.

## Execution plan

| | |
|---|---|
| **Now** | MVP live on GIWA Sepolia: two products, physical settlement, maker accounts, request for quote, public faucets and the first live position. |
| **Phase 2** | External makers onboarded against the published quote API; RedStone as a second oracle behind the same adapter. |
| **Phase 3** | Bridged USDC and wrapped BTC settlement, multisig ownership and KMS-held keys, mainnet readiness. |
| **Phase 4** | KRW-denominated strikes when a KRW stablecoin lands on GIWA. |

## Counterparty model and honest status

Counterparties are market makers, not the protocol and not other users. Premiums are paid from a maker's own on-chain balance, and the contract refuses a trade that maker could not deliver on. At expiry only that position's own collateral and reserve move. Writer collateral is never pooled with another writer's.

On testnet the maker is operated by the team, which is stated plainly rather than dressed up as liquidity. The architecture is the multi-maker one either way: registering a second maker is three calls, and the request-for-quote flow already selects the best bid among however many answer.

**Status: testnet, unaudited, mainnet pending GIWA mainnet.**